

QUICK ACTIVITY 1

1. Identify the inputs, processes, and outputs.

Input	Processes	Output
Number of notebooks	Calculate notebook cost	Total cost
Number of pens	Calculate pen cost	change
Payment amount	Calculate total cost and change	

2. Divide the problem into smaller tasks. Write the tasks in their correct order.

- Start the program.
- Input the number of notebooks.
- Input the number of pens.
- Input the payment amount.
- Calculate the total cost of notebooks.
- Calculate the total cost of pens.
- Add both costs to get the total cost.
- Subtract the total cost from the payment to get the change.
- Display the total cost and change.
- End the program.

3. Present your solution using either pseudocode or a flowchart.

START

```
READ notebooks, pens, payment
notebookCost ← quantity × ₱45
penCost ← quantity × ₱12
totalCost ← notebookCost + penCost
change ← payment – totalCost
Display totalCost, change
```

END

4. Perform a dry run using both test cases below. Show each calculation and the resulting values.

Step	Variable	Calculation	Result
1	notebookCost	$3 \times \text{₱}45$	₱135
2	penCost	$2 \times \text{₱}12$	₱24
3	totalCost	$\text{₱}135 + \text{₱}24$	₱159
4	change	$\text{₱}200 - \text{₱}159$	₱41

Total cost: ₱159

Change: ₱41

Step	Variable	Calculation	Result
1	notebookCost	$2 \times \text{₱}45$	₱90
2	penCost	$3 \times \text{₱}12$	₱36
3	totalCost	$\text{₱}90 + \text{₱}36$	₱126
4	change	$\text{₱}150 - \text{₱}126$	₱24

Total cost: ₱126

Change: ₱24